

L1.7a Implicit Differentiation

Work in your group. Check your answers against the Quarto tonight.

Implicit curves and level sets

1. Let $S(x, y) = x^2 + y^2$, so the circle $x^2 + y^2 = 25$ is the curve $S = 25$. Start at $(3, 4)$.
 - (a) Take the step $dx = -0.1$, $dy = -0.1$. Find dS , and find the level you land on. **Are you still on the circle?**
 - (b) Now take $dx = 0.1$ and find the dy that makes $dS = 0$. Then write down $\frac{dy}{dx}$. **What have you just found?**

Implicit differentiation

2. Differentiate each with respect to x .

$$\frac{d}{dx}(y^3)$$

$$\frac{d}{dx}(xy)$$

$$\frac{d}{dx}(\sin y)$$

$$\frac{d}{dx}(x^2y)$$

Why does a y pick up a y' and an x does not?

3. Take $\frac{d}{dx}$ of both sides. **Stop there — do not solve for y' .**

(a) $x^2 + y^2 = 25$

(b) $x^3 + y^3 = \frac{9}{2}xy$

(c) $y + x \cos y = x^2$

4. Now finish the job — find y' .

(a) $x^2 + y^2 = 25$

(b) $x^2 + xy + y^2 = 3$

(c) $y + x \cos y = x^2$

Tangent lines to an implicit curve

5. Find the equation of the tangent line.

(a) $x^2 + y^2 = 25$ at $(3, 4)$ (compare with the entry ticket)

(b) $x^3 + y^3 = \frac{9}{2}xy$ at $(2, 1)$ (check the point is on the curve first)

Horizontal and vertical tangents

6. Use your y' from item 4(b). On $x^2 + xy + y^2 = 3$, find **every** point with a horizontal tangent and **every** point with a vertical tangent.

7. **Stretch.** Where on $x^2 + y^2 = 25$ is the tangent line **parallel to** $y = x$?

8. **BOARDS** The ellipse $2x^2 + y^2 - 4x + 2y + 2 = 0$ is graphed below. Find **every** horizontal tangent and **every** vertical tangent.

